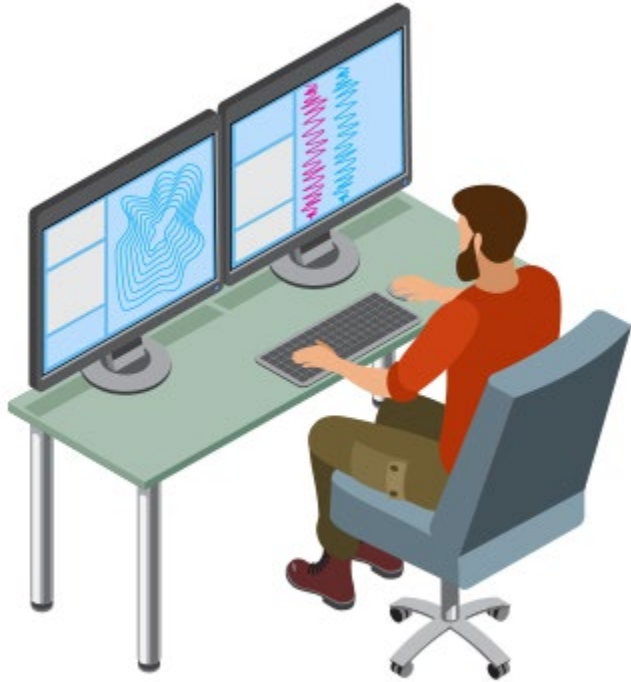




# Getting Started with Geographic Information Systems (GIS)

Intro to GIS + Map Creation Basics

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# What is GIS?

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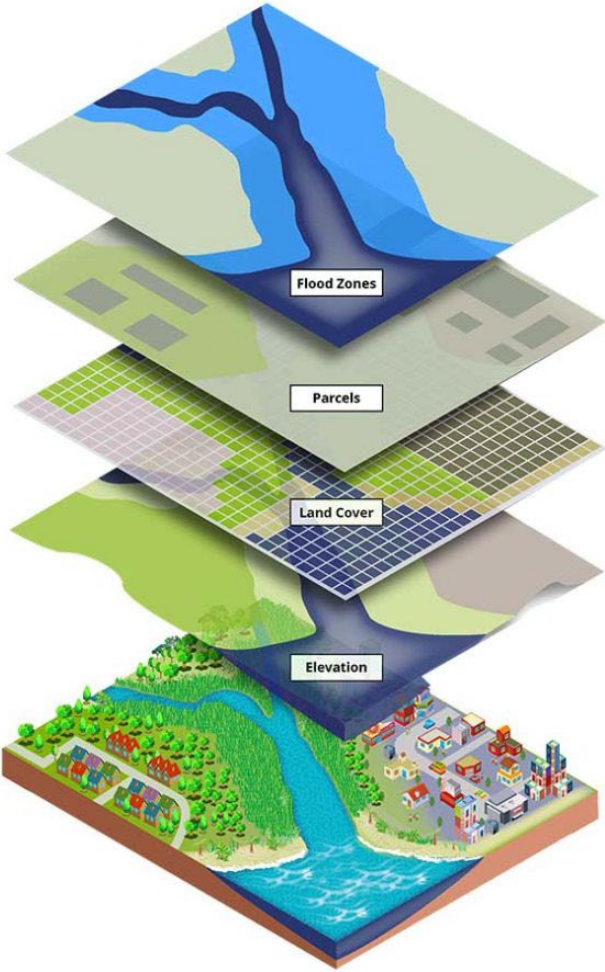
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## Analyze

Discover patterns and relationships

## Visualize

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# What Makes GIS Powerful?

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## How do places relate?

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## What patterns exist?

Reveal trends invisible in spreadsheets.



## What has changed?

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## Spatial Data — Where

-   $\frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$

- **Name** — label or identifier
- **Type** — category or classification
- **Population** — numeric values
- **Status** — condition or state
- **Any description** — additional context

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Data Layers

Your custom geographic information overlaid on the basemap.



Symbology

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Pop-ups

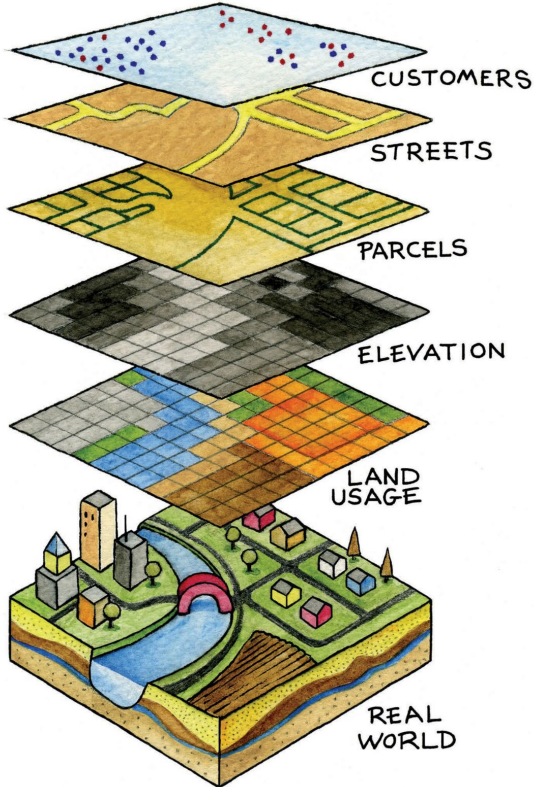
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# What is a Map in GIS?



A GIS map is **more than a static image** .  
It's a dynamic collection of **layers** —  
each holding a different dataset that  
can be turned on or off, styled, and  
analyzed.

**i** Think of it like a stack of  
transparent sheets. Each  
sheet represents a different  
dataset — roads, buildings,  
parks — layered together to  
tell a complete story.

# Getting Started

01

What is a map?

A map is a visual representation of a geographic area. It can show physical features like mountains and rivers, or human-made features like cities and roads. Maps can be used for a variety of purposes, including navigation, education, and research.

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Why use a map?

Maps are used to show the location of places, to understand the spatial relationships between different features, and to make decisions about how to use land. Maps are also used to communicate information about the world to a wider audience.

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Check your map and draw conclusions from what you see.

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Find or Load Your Data

CSV files, GIS servers, open data portals, or institutional datasets.

05

Style Your Layers

Apply colors, symbols, and sizes to represent your data.

03

Choose a background reference map.

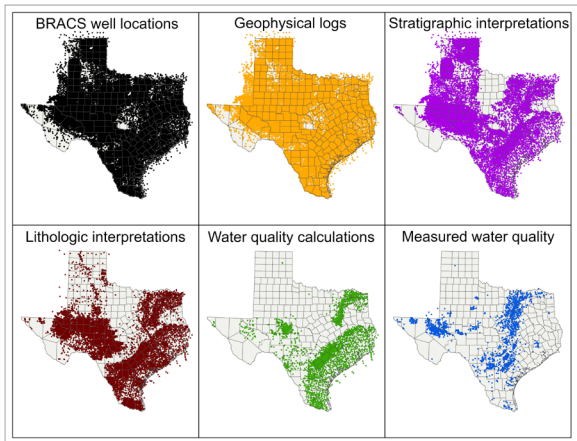
Choose a background reference map.

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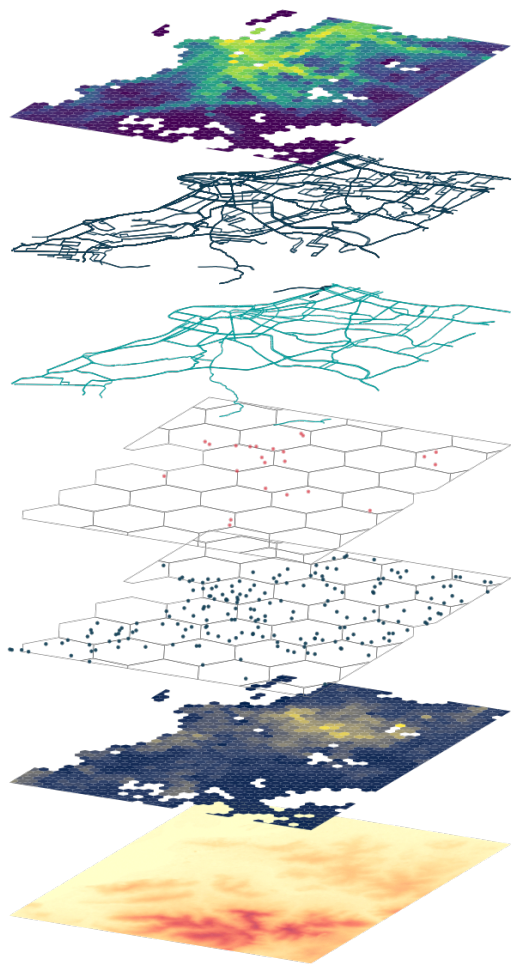
Configure Pop-ups

Pop-ups are used to display additional information about a specific location on a map. They can be used to show data from a database, to display a description of a place, or to provide a link to a website.





📄 **Example questions:** Where are campus resources? How are services distributed? Where are environmental risks highest?



Accessibility

Public transport

Roads

Hospitals

Schools

Population

Terrain

## Step 2: Add Your Data

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Spreadsheets (CSV)

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Open Data Portals

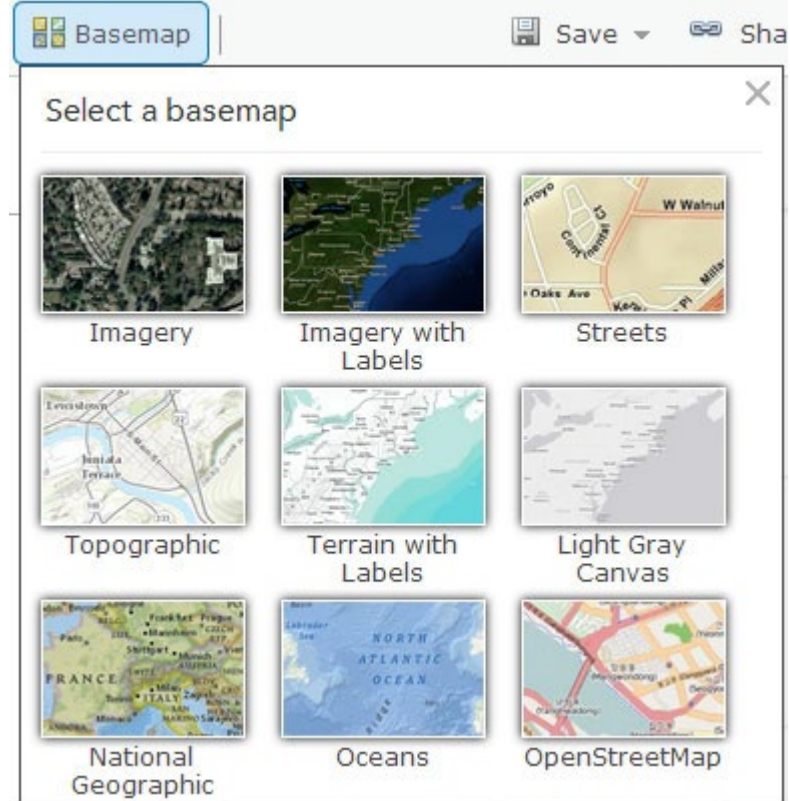
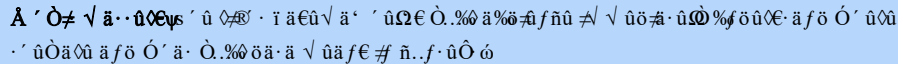
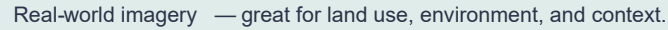
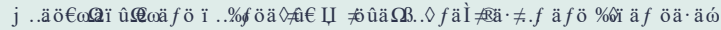
Public datasets from governments and organizations



Institutional Datasets

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| Value |        |       |       |
| Size  |        |       |       |

## Design with Purpose

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Goal: Make patterns easy to understand without reading a legend.



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You now have the core building blocks of a functional GIS map. Here's a quick recap of your progress:

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[illegible] $\frac{7}{8}$ 

## Basic Styling Applied

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Launch the GIS mapping platform.

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